Sameek Bhattacharya

Texas, USA | sameekbhattacharya@my.unt.edu | LinkedIn | Website | GitHub

➢ Education

University of North Texas, Denton, TX *(Expected)* **May-2026**
Masters(Thesis) - Artificial Intelligence
University of Engineering and Management, Kolkata, WB, India **Jun-2021**
Bachelors - Computer Science and Engineering

➢ Skills

Programming Languages: Python, Java, C, HTML, CSS, C++, JavaScript
Research Skills: Adversarial Attacks, Deepfake detection, Unimodal and multimodal data processing (audio and images)
Data Science & Machine Learning: Audio Processing, DeepFake detection, Image Processing, Computer Vision, Feature Engineering, Natural Language Processing, Gaussian Naive Bayes, k-Nearest Neighbor
Cloud Technologies: Kaggle, GoogleWorkspace(Colab, Slides), GitHub, Atlassian (Bitbucket)
DevOps & Version Control: Anaconda, Git
Other Tools & Technologies: PyTorch, TensorFlow, Jupyter Notebook, VS Code, Atlassian (Jira, Confluence), Adobe Illustrator, Adobe Effects, Regression Testing, Agile Framework, InShot

➢ Research Experience

University of North Texas **Denton, Texas**
Research Assistant at the Visual Computing and Biometric Security Lab *Feb 2025 - Current*

- **Project Aims & Datasets:** Focused on evaluating model robustness and developing novel GAN-based adversarial attacks using **AudioSet** (7 classes), **ASVspoof2019** (48 speakers), and **DCASE2019 Task 1** datasets.
- **Architecture & Preprocessing:** Constructed detection architectures utilizing **ResNet50** with Fully Connected layers and **Xception** with Logistic Regression, extracting **STFT features** from raw **.wav** files via **librosa** and saving them as **NumPy** arrays.
- **Adversarial Implementation:** Challenged the baseline with **Gaussian noise** (factors: 0.05, 0.005) and white-box attacks including **FGSM** (epsilon: 0.01, 0.001, 0.0001), **BIM** & **PGD** (epsilon: 0.0001, 0.00001, 0.000001), **DeepFool** (overshoot: 0.02, 0.0001, 0.00001), and **C&W** (c: 10, 1, 0.1, 0.01, 0.001).
- **Attack Results:** Achieved a wide Attack Success Rate (ASR) range of **4% to 99%** depending on the specific attack parameters, while generating attacked spectrograms for visual analysis of the perturbations.
- **GAN Baseline Performance:** Currently developing a **PANN-CNN14** baseline model for GAN-based attacks on Audio Event Classification, which has reached **96% validation accuracy** and **66% test accuracy** over **DCASE2019** dataset.
- **Future Objectives:** Aiming to achieve high ASR with GAN-based techniques to demonstrate the failure of existing defenses, followed by the development of a novel defense mechanism to counter these generative attacks.

➢ Work Experience

University of North Texas **Denton, Texas**
Teaching Assistant (for Computer Science I) *Aug 2025 - Current*

- Enhanced student learning outcomes by supporting the **professor** in developing **course materials**, leading **lab sessions each of 30+ students**, grading **assessments** with constructive feedback, and providing individualized guidance on core **c++ programming concepts**.

Graduate Research Assistant *May 2025 - July 2025*

- Led a 6-week state-funded **Learn AI Summer Camp** in Texas for **35+ high school students** weekly, delivering hands-on instruction in **Python programming** (basic to intermediate), **K210 Developer Kit**, and **XGO Robot Dog with micro:bit** via **Microsoft MakeCode**, while also conducting data analysis of **student demographics** and feedback through registration forms and pre/post-camp surveys to improve program outcomes.

Institute of Engineering and Management **West Bengal, India**
Research Assistant *Jan 2023 - Oct 2023*

- Developed **AI-driven tools** including a **life-cycle management framework** with **results dashboard** and a **benchmarking system** for **tabular and image datasets** using **Scikit-learn** and **TensorFlow**.
- Analysed **student performance data** to provide actionable insights to the **education board** and oversaw **academic operations** such as **grade card publication**, **transcripts**, **certificates**, and **graduation ceremonies**.

TATA Consultancy Services **West Bengal, India**
System Engineer *Aug 2021 - Apr 2022*

- Developed and launched **Johnson & Johnson Institute's training website** for doctors and staff, working as a **full-stack web developer** using **HTML**, **CSS**, **Bootstrap**, **JavaScript**, **Git**, and **Atlassian tools**.
- Ensured **compliance** with **Johnson & Johnson standards** by completing certifications in **iQMS**, **HSE**, **Data Privacy**, and **Health & Safety**.

➢ Community Service Experience

Kolkata Robotic Society — **West Bengal, India**

Co-Founder — *Sept 2017 - May 2020*

- Co-founded a **non-profit educational organization** to promote **tech awareness** by teaching **high school students AI**, **machine learning**, **robotics**, **Arduino**, and **drones**, reaching 150+ participants through approx. 20 sessions and a **state-level workshop**.
- Developed **entrepreneurial skills** by engaging with **investors** and **entrepreneurs** to support and grow the organization.

➢ Publications *(International Conference)*

Bhattacharya, Sameek, et al. 2021, December. Finding a baseline machine learning model in Sci-Kit Learn package in terms of accuracy and efficiency. In IEEE Pune Section International Conference (PuneCon), pp. 1-7. IEEE. 2021. India. DOI: 10.1109/PuneCon52575.2021.9686536

➢ Project Experience

- Engineered a wearable system using **ESP32, INA219, UV sensor** and **solar panel** to correlate power output with UV intensity across **~5,000 samples**, achieving **97% accuracy** (**Logistic Regression**) and **96%** (**SVM**) in classifying high UV exposure.
  - The dataset is split into **70% training, 15% validation, and 15% testing**, ensuring the model is evaluated on data it has never seen before. Evaluation metrics **Accuracy, Precision, Recall, F1 score**.
  - **Solar Panel matched the pattern of UV sensor readings**. The solar panel integrates UV detection, cutting costs while delivering a self-sustaining, lightweight device.
  - **Logistic Regression** model performs best giving **97% accuracy** and **SVM** gave around **96% accuracy**. The **baseline** model reached **93% accuracy**.
- Developed a lightweight **Automated Essay Scoring (AES)** system using a statistically determined sample of **141 essays** and **53 optimized features** (including top **50**-word counts), achieving a near-perfect **Quadratic Weighted Kappa** of **0.9595** with a **Gradient Boosting Regressor**, proving that **simple regression models** can effectively score essays in **low-resource settings**.
  - Conducted a statistically rigorous Automated Essay Scoring (AES) study using a carefully selected dataset of **141 essays** (exceeding the calculated minimum of **113**), reducing an initial **1,221 features** to **50 top word-count predictors** via **Univariate Feature Selection**.
  - Achieved a near-perfect **Quadratic Weighted Kappa (QWK)** score of **0.9595** using a **Gradient Boosting Regressor**, significantly outperforming classification models like **KNN** (**QWK 0.6266**) and demonstrating that essay scoring is best treated as a **regression task**.
  - Validated that a lightweight, interpretable model using only **53 optimized features** (including unique word and stopword counts) can approximate human grading in **low-resource settings**, with regression models consistently achieving **lower error rates** (**RMSE 0.9264**) than classifiers.
- Developed a **Reddit comment engagement predictor** using the **Kaggle 1M Reddit Comments dataset**, classifying comments by **engagement level** and **sentiment** using **TF-IDF with Logistic Regression** and **RoBERTa** models by subsampling 30k samples from **balanced and imbalanced** subsets.
  - Fine-tuned **RoBERTa** for sentiment analysis, achieving superior classification with **F1-scores of 0.65 for positive**, **0.47 for neutral**, and **0.11 for negative** sentiment on the 30k dataset.
  - Evaluated **TF-IDF + Logistic Regression baseline**, analysing its performance on both balanced (**F1-scores: 0.41 Negative**, **0.39 Neutral**, **0.40 Positive**) and imbalanced data (**F1-scores: 0.11 Negative**, **0.45 Neutral**, **0.54 Positive**) to demonstrate the advanced model is significant.
- Built and compared **skin cancer classification model** using the **ISIC dataset**, distinguishing **benign** and **malignant** classes.
  - Did a comparative analysis of four CNN architectures (**ResNet50, DenseNet121, VGG16, InceptionV3**) for **skin cancer classification**, evaluating the impact of preprocessing steps like **CLAHE** and **polynomial (x^2) feature transformation**.
  - Achieved a top **F1-score of 0.89** for both **benign** and **malignant** classes using **DenseNet121** with **CLAHE**, demonstrating its superior and balanced performance over other models like **ResNet50 (F1: 0.88 - benign/0.89 - malignant)** and **VGG16 (F1: 0.87 - benign/0.87 - malignant)** in this specific **F1-score**.
- Implemented **Gaussian Naïve Bayes from scratch**, achieving **86% accuracy** on the **Scikit-learn digit recognition dataset** (1797 grayscale 8×8 images).
- Designed and trained **K-Nearest Neighbors (KNN) from scratch**, tuning hyperparameters (**k = 1, 3, 5, 100, 500**) and achieving **99% accuracy at k=3** on the same dataset.
- Led a team of 4 as **Project Manager** to develop a website for the **Kaggle – Spaceship Titanic** competition.
  - Integrated **Logistic Regression**, achieving **62.22% accuracy**, **0.6156 F1-score** and **0.6748 AUC-ROC** and **CatBoost**, achieving **79.79% accuracy**, **0.7914 F1-score** and **0.8611 AUC-ROC**, to predict passenger survival outcomes.
- Worked on multiple self-initiated projects involving **Computer Vision** and **Robotics (Arduino)**:
  - Applied **OpenCV** to process real-time webcam captures into **grayscale**, **inverted**, and **sepia** views.
  - Used **Haar Cascade classifiers** for face detection from captured images.
  - Designed and programmed robots using **Arduino Uno**, including **Line Follower bots**, **Obstacle Avoider bots**, **manual multi-terrain bots**, and **war-robots** for robotics competitions across India.